Improvement of the eTOM Operations phase through the comparison with ITIL best practices

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***Abstract* — IT services, daily management and daily use of those services is nowadays part of every business. That's why there are many standards to define management and control of use for IT services. One of them is eTOM standard which represents a most common standard for building and management of IT services in telecommunication industry, like Telecom operators. Another one is ITIL which can be used in every industry and is very common in use. The main goal of this paper is to suggest new improvement of eTOM standard based on those components and characteristics of ITIL that current eTOM standard doesn't have. Idea for improvement is based of comparison of Service Desk implementation using eTOM standard and Service Desk implementation using ITIL best practices wile Service Desk in both cases has same goal of ensuring functionality and availability of services and stable communication with users.**

***Keywords* — eTOM (enhanced Telecom Operations Map), ITIL (Information Technology Infrastructure Library).**

I. Introduction

IT managers are challenged to supervise and manage IT services and projects While planning, projecting and managing any IT service or process, IT manager needs to use standards because of the given ideas and options and best practices described inside, but every decision has to be according to plans and goals of the organization. There are few best known standards used for IT Service Management and they are: ITIL, eTOM, COBIT, PRINCE2, Learn and IBM Tivoli Unified Process (ITUP).

ITIL is a collection of best practices for managing IT services in organization of any type. It consists of 5 phases: Service Strategy, Service Design, Service Transition, Service Operation and Continual Service Improvement. These phases include together 23 processes and 4 functions. There is to restriction about the type of business to which managers can apply ITIL [2].

eTOM (enhanced Telecom Operations Map) is standard for managing IT services in Telecommunication industry. It consists of 3 phases: strategic, operational and management. These phases include together 15 processes[1]. Although eTOM is standard specialized in telecommunication industry, it's application is spreading in other types of organizations.

Service Desk is IT service and it's goal is to provide point of contact to meet communication needs for external users of service and for other IT employees, but it needs to be according to defined rules and agreement between customer and IT provider.

The main goal of this paper is to improve the existing eTOM standard through a comparison with the ITIL best practices and the implementation of Service Desk function using these two standards. Later in the paper, suggested improvement of eTOM standard will be presented based on ITIL but concluded on Service Desk implementation through both standards. The idea of suggested improvement is to enable better ideas and opportunities for managers when designing and developing business processes in organization especially Service Desk service. The second section of the paper gives a short list of previous researches in this area. The third section of the paper describes the implementation of Service Desk service in an organization using eTOM standard, while fourth section describes the implementation of Service Desk service using ITIL best practices. The fifth section of the paper represents the comparison of these two implementations of Service Desk and as a result provides a new improved eTOM standard. In the conclusion of the paper are presented eventual improvement which may be achieved by using new improved eTOM standard in implementation of IT services.

Service Desk offers these services: forum on the organization's web site where registered users can leave comments or report a problem or ask a question, chat for private communication between users and Service Desk employees, phone calls and email communication for help or reporting a problem or comments. But this services are not just for external users, organization employees can use them too.

II. Previous Research

This paper represents the continuity of researches of authors to produce a new universal IT Service Management framework which could replace a today's the most popular framework which is called ITIL [12], [13], [14], [15], [16].

The most interested paper connected to this paper is [17]. The aim of this paper is to see in which ITIL processes the result of measurement is bad, to find complementary ISO/IEC 20000 processes in which the result is good, and based on this to suggest a new model of ITIL framework for the design and implementation of the Billing system for x-play services of Telecom operator. The scientific value of this paper is a new produced ITIL

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framework which could be used also for some other Telecom operator's systems.

The second interesting paper for the research covered in this paper is [18]. The goal of this paper is to introduce a new and improved model for eTOM standard through the results of an empirical study in students, regarding quality of services Telecom operators provide. The emperical survey took place at the University of Ioannina in Greece. The results showed certain disadvantages in the current business processes of Telecom operators. In conclusion, the paper presents a new and improved proposal for eTOM standard which includes new processes from previous done research. The paper gives as the result three new processes: New IP Services Development & Management (Strategy phase), Information Security Development & Management (Strategy phase) and Service Desk Management (Operations phase).

III. Implementation of Service Desk According to eTOM Standard

Because of the structure of eTOM standard Service Desk as a service is part of Assurance vertical process group which intersects with four horizontal process groups: Customer Relationship Management, Service Management and Operations, Resource Management and Operations, Supplier/Partner Relationship Management. Every process group has some specific incidents and problems[1], [3].

Customer Relationship Management is defining the quality of a service and is making decision if solvation of one problem needs to be transferred to other process group of Assurance phase. Therefore, Customer Relationship Management is actually managing other three process groups. Service Desk needs to develop a plan of work, a strategy of giving support to users and of evaluating the results of work and resolved incidents or problems but everything needs to be according to defined rules and goals [1], [3], [4].

Service Management and Operations focuses on the knowledge of services and includes all functionalities necessary for the management and operations of communications and information services required by or purposed to customers. Service Desk needs to be able to see all the knowledge and to understand the connections so appropriate decision could be made in case of problems or incidents and to distinguish who should resolve the situation [1], [3], [4].

Resource Management and Operations maintains the knowledge of the resources and is responsible for management of all resources needed for delivery and support of the services to users. Also, it includes all functionalities for direct management of those resources. Service Desk needs to have access to be aware which resources are assigned to which services especially when the problem or the incident occurs [1], [3], [4].

Supplier/Partner Relationship Management supports the core operational processes. Service Desk needs to know who are partners and suppliers in every situation and are they responsible for incurred problem or incident and should they resolve it [1], [3], [4].

This way problem or incident flows through different layers of Service Desk service inside the organization or even to external suppliers or partners.

IV. Implementation of Service Desk According to ITIL Standard

In case of Service Desk implementation according to ITIL standard, Service Desk as a service as part of Operation phase. So because of that there is five important processes in Operation phase that are very valuable for Service Desk [5].

Event Management is in charge for detection, classification and specification of every situation reported to Service Desk employees. This way situation can be resolved before it progresses into a problem or incident. Very positive side of this that it decreases costs for organization that is giving the service and to the customers using the service. Every detected event is stored in a database so it could be used for comparison in the case of new situation[5], [7].

Incident Management is focused on detection, classification, categorization, defining priorities, diagnostics, escalation and finding solution of reported incidents. Closure of resolved incident includes saving the data about incident into database so the performance of Service Desk employees could be increased and maybe better solution for the incident could be found[5], [9].

Request Fulfillment is intended for realizing requests for smaller changes or requests for some information. There are many ways for customers to send requests to Service Desk employees. Requests get priority by their importance for the system or service or by the risk level that exists while request is not implemented or by time needed for its implementation [5], [11].

Access Management is in charge for defining and granting access to users and employees so integrity, availability and confidentiality of information and services that the organization is offering. Service Desk employees can give or take more or less rights to defined groups of users and employees. This way it’s much easier to manage and monitor people, documents and changes and much easier to see failure or malversation [5], [10].

Problem Management is focused on detection, classification, categorization, prioritizing, diagnostics and solving the problems. Closure of resolved problem includes saving the data about the problem into database so employees of Service Desk could use that data for future situations and maybe improve existing solution [5], [8].

V. Implementation of New eTOM Standard

New improved eTOM standard is based on the ITIL standard processes which currently do not exist in the eTOM standard. That way planning and managing of Service Desk service in organization can be done with a new perspective and enable improvement of performance and decrease the costs.

Based on comparative analysis of Service Desk implementation using eTOM standard and Service Desk implementation using ITIL best practices, it can be concluded that Service Desk implementation using eTOM standard is missing Event management as a process. This process is very important and is enabling better Service Desk service and is improving communication between users and Service Desk employees by giving faster and more efficient help before event grows into incident or problem [1], [2], [6].

Looking at the structure of eTOM standard next to ITIL best practices, it can be concluded that Service Operation phase in ITIL can be mapped as a combination of Fulfillment, Assurance and Billing process groups in eTOM. Because Event Management is part of Operation phase, Service Desk employees are able to react more appropriate in reported situations by using information gathered from previous events and even maybe find new and better solution of one event is reported few times with same symptoms. This process currently is not part of eTOM standard but through this paper it's suggested for this change, because this small improvement of eTOM standard is making big improvement of Service Desk as a service. It is important not only to employees but to users because every situation reported can be resolved before it becomes more serious and more expensive. While in current eTOM standard users are reporting the situation when it's already an incident or a problem. This way it might take too much time, too much resources, too much money not only for users but for organization itself and its employees [1], [2], [3], [5].

Main difference between Service Operation phase in ITIL and Assurance process group in eTOM is that Service Operation includes process with main function to implement and fulfill requests from users. This process is Request Fulfillment. In eTOM standard this process is not part of Assurance but it is a self-standing process group called Fulfillment and it has the same goal as Request Fulfillment in Service Operation. For easier comparison, here are overlaps of eTOM Assurance processes with ITIL Service Operation processes.

TABLE 1: ETOM ASSURANCE OVERLAPS WITH ITIL SERVICE OPERATION[1]

eTOM Assurance /ITIL Service Operation	EM	IM	PM	RF	AM
CRM	+			+	+
SM&O	+	+	+		+
RM&O	+	+	+		+
S/PRM				+	+

Event Management overlaps with Customer Relationship Management because customers are usually the one reporting events and because interaction with customers is very important for understanding their needs and gathering more information.

Event, Incident and Problem Management overlap with Resource Management and Operations and Service Management and Operations because reported events, incidents and problems can be about services or resources.

Access Management overlaps with Service and Resource Management and Operations and Service Management and Operations and Supplier/ Partner Relationship Management because access rules to organization's system, resources and services need to be defined for employees of the organization, for partners, suppliers and for final users.

Request Fulfillment overlaps with Customer Relationship Management and Supplier/ Partner Relationship Management because requests are not just coming from customers but from and to partners and suppliers and usually are defined in contracts.

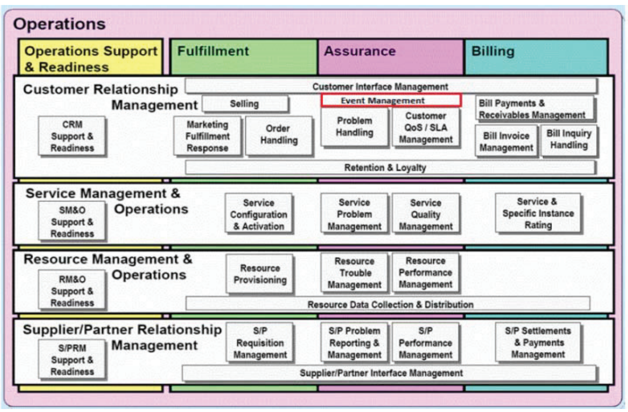
Next picture presents the Operations process group of eTOM standard with additional process as an improvement, Event Management in Assurance part.

As it can be seen on the Picture 1, Event Management is positioned in Customer Relationship Management because every reported situation comes from customers, external or internal.

VI. RESULTS OF COMPARISON

Using improved eTOM standard to implement Service Desk as a service in an organization, there is no need to change the structure of Service Desk but only to reorganize the schedule of taking and working on reported situations then resolving them, saving data about them and telling the user about the result, This time these reported situations are mostly events that are like warnings about coming incidents and problems. This way, employees can analyze events to conclude about potential incidents and problems, to make proactive decisions, to prepare for actions in case it turns into incident or problem earlier than expected. That means that employees are able to plan the strategy for solving reported situations and to help each other if needed.

Improved eTOM standard used in organization of any type enables new options for managers while analyzing, planning and managing services.



PICTURE 1: IMPROVED ETOM STANDARD

It's important for Service Desk employees to have strategy for working with events. So it would take less time, less resources and less money to resolve it, to

[1] CRM is Customer Relationship Management,
SM&O is Service Management and Operations,
RM&O is Resource Management and Operations,
S/PRM is Supplier/Partner Relationship Management,
EM is Event Management,
IM is incident Management,
PM is Problem Management,
RF is Request Fulfillment and
AM is Access Management

improve service if needed, to change some part if needed or to forward event to next level or to supplier or partner if it's up to elements that they are providing.

PICTURE 2: IMPROVED ASSURANCE PROCESS GROUP IN ETOM

Event Management is very useful because resolving smaller and easier situations gives more opportunities for employees to create better communication with customers, with suppliers and partners. It also gives more feedback from customers and a chance for employees to become more familiar with customer's systems and their needs. This way Service Desk employees can create structured plan for accepting reported events, their analysis, prioritizing, forwarding and resolving but saving data about the event for further work and improvement.

Using Event Management process decreases need for changing system components and for major changes on systems, because resolving events disables problem and incident appearance.

VII. CONCLUSION

In the end, it's possible to conclude that new improved eTOM standard will allow increased information usage for Service Desk employees, increase support level for end-users and improve the level of control over systems at end users and over services and system in the organization. If comparison is made between old and new version of eTOM standard, this is what the new one offers:

- Proactive activities
- Lower costs of resolving issues
- Lower costs of upgrading the systems
- Suppression of incidents and problems development
- Structured database about all reported issues
- Better communication between employees
- Better communication with end-users
- Less time needed for resolving issues
- More control over system
- Detailed analysis of reported issue
- Detailed analysis of end-user requests
- Working on smaller system components.

Future research in this area will be focused on connecting the new and improved eTOM standard with other standards for managing and planning services inside an organization. Also other idea is to analyze and compare implementation of different services by using different standards with a goal of their mutual work.

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